

$$\begin{aligned}
C_{\text{UB}\phi_1}^{11,12} \rightarrow & \frac{1}{16\pi^2} \left(\frac{1}{432} g_1^3 y_u^{11i2} \text{LF}_{2,1,0} [m_1, m_q^{i1}] - \frac{1}{216} g_1^3 y_u^{i1i2} \text{LF}_{3,1,-1} [m_1, m_{\bar{q}}^{i1}] + \right. \\
& \frac{1}{432} g_1^3 y_u^{i1i2} \text{LF}_{4,1,-2} [m_1, m_q^{i1}] + \frac{4}{27} g_1^3 y_u^{i1i2} \text{LF}_{2,1,0} [m_1, m_{\bar{u}}^{i2}] - \\
& \frac{8}{27} g_1^3 y_u^{i1i2} \text{LF}_{3,1,-1} [m_1, m_{\bar{u}}^{i2}] + \frac{4}{27} g_1^3 y_u^{i1i2} \text{LF}_{4,1,-2} [m_1, m_{\bar{u}}^{i2}] + \\
& \frac{1}{16} g_1 g_2^2 y_u^{i1i2} \text{LF}_{2,1,0} [m_2, m_{\bar{q}}^{i1}] - \frac{1}{8} g_1 g_2^2 y_u^{i1i2} \text{LF}_{3,1,-1} [m_2, m_q^{i1}] + \\
& \frac{1}{16} g_1 g_2^2 y_u^{i1i2} \text{LF}_{4,1,-2} [m_2, m_{\bar{q}}^{i1}] + \frac{1}{9} g_1 g_3^2 y_u^{i1i2} \text{LF}_{2,1,0} [m_3, m_q^{i1}] - \\
& \frac{2}{9} g_1 g_3^2 y_u^{i1i2} \text{LF}_{3,1,-1} [m_3, m_{\bar{q}}^{i1}] + \frac{1}{9} g_1 g_3^2 y_u^{i1i2} \text{LF}_{4,1,-2} [m_3, m_q^{i1}] + \\
& \frac{4}{9} g_1 g_3^2 y_u^{i1i2} \text{LF}_{2,1,0} [m_3, m_{\bar{u}}^{i2}] - \frac{8}{9} g_1 g_3^2 y_u^{i1i2} \text{LF}_{3,1,-1} [m_3, m_{\bar{u}}^{i2}] + \\
& \frac{4}{9} g_1 g_3^2 y_u^{i1i2} \text{LF}_{4,1,-2} [m_3, m_{\bar{u}}^{i2}] - \frac{1}{12} g_1 \overline{y_d^{\text{pr}}} y_d^{i1r} y_u^{pi2} \text{LF}_{2,1,0} [\tilde{\mu}, m_d^r] + \\
& \frac{1}{24} g_1 \overline{y_d^{\text{pr}}} y_d^{i1r} y_u^{pi2} \text{LF}_{3,1,-1} [\tilde{\mu}, m_d^r] + \frac{1}{24} g_1 \overline{y_d^{\text{pr}}} y_d^{i1r} y_u^{pi2} \text{LF}_{4,1,-2} [\tilde{\mu}, m_d^r] + \\
& \frac{1}{12} g_1 \overline{y_u^{\text{pr}}} y_u^{pi2} y_u^{i1r} \text{LF}_{2,1,0} [\tilde{\mu}, m_q^p] - \frac{5}{12} g_1 \overline{y_u^{\text{pr}}} y_u^{pi2} y_u^{i1r} \text{LF}_{3,1,-1} [\tilde{\mu}, m_q^p] + \\
& \frac{1}{3} g_1 \overline{y_u^{\text{pr}}} y_u^{pi2} y_u^{i1r} \text{LF}_{4,1,-2} [\tilde{\mu}, m_q^p] + \frac{1}{6} g_1 \overline{y_u^{\text{pr}}} y_u^{pi2} y_u^{i1r} \text{LF}_{2,1,0} [\tilde{\mu}, m_{\bar{u}}^r] - \\
& \frac{5}{24} g_1 \overline{y_u^{\text{pr}}} y_u^{pi2} y_u^{i1r} \text{LF}_{3,1,-1} [\tilde{\mu}, m_{\bar{u}}^r] + \frac{1}{24} g_1 \overline{y_u^{\text{pr}}} y_u^{pi2} y_u^{i1r} \text{LF}_{4,1,-2} [\tilde{\mu}, m_{\bar{u}}^r] - \\
& \frac{5}{108} m_1 g_1^3 a_u^{i1i2} \text{LF}_{2,1,1,0} [m_1, m_q^{i1}, m_{\bar{u}}^{i2}] + \frac{1}{72} m_1 g_1^3 a_u^{i1i2} \text{LF}_{2,2,1,-1} [m_1, m_{\bar{q}}^{i1}, m_{\bar{u}}^{i2}] + \\
& \frac{5}{108} m_1 g_1^3 a_u^{i1i2} \text{LF}_{3,1,1,-1} [m_1, m_{\bar{q}}^{i1}, m_{\bar{u}}^{i2}] - \frac{1}{72} g_1^3 y_u^{i1i2} \text{LF}_{1,1,1,0} [m_1, m_{\bar{q}}^{i1}, \tilde{\mu}] + \\
& \frac{1}{36} g_1^3 y_u^{i1i2} \text{LF}_{2,1,1,-1} [m_1, m_q^{i1}, \tilde{\mu}] - \frac{1}{72} g_1^3 y_u^{i1i2} \text{LF}_{3,1,1,-2} [m_1, m_q^{i1}, \tilde{\mu}] - \\
& \frac{1}{72} m_1 g_1^3 a_u^{i1i2} \text{LF}_{2,2,1,-1} [m_1, m_{\bar{u}}^{i2}, m_{\bar{q}}^{i1}] + \frac{2}{9} g_1^3 y_u^{i1i2} \text{LF}_{1,1,1,0} [m_1, m_{\bar{u}}^{i2}, \tilde{\mu}] - \\
& \frac{4}{9} g_1^3 y_u^{i1i2} \text{LF}_{2,1,1,-1} [m_1, m_{\bar{u}}^{i2}, \tilde{\mu}] + \frac{2}{9} g_1^3 y_u^{i1i2} \text{LF}_{3,1,1,-2} [m_1, m_{\bar{u}}^{i2}, \tilde{\mu}] - \\
& \frac{5}{144} g_1^3 y_u^{i1i2} \text{LF}_{2,2,1,-2} [m_1, \tilde{\mu}, m_q^{i1}] + \frac{5}{36} g_1^3 y_u^{i1i2} \text{LF}_{2,2,1,-2} [m_1, \tilde{\mu}, m_{\bar{u}}^{i2}] + \\
& \frac{1}{8} g_1 g_2^2 y_u^{i1i2} \text{LF}_{1,1,1,0} [m_2, m_{\bar{q}}^{i1}, \tilde{\mu}] - \frac{1}{4} g_1 g_2^2 y_u^{i1i2} \text{LF}_{2,1,1,-1} [m_2, m_{\bar{q}}^{i1}, \tilde{\mu}] + \\
& \frac{1}{8} g_1 g_2^2 y_u^{i1i2} \text{LF}_{3,1,1,-2} [m_2, m_q^{i1}, \tilde{\mu}] + \frac{5}{16} g_1 g_2^2 y_u^{i1i2} \text{LF}_{2,2,1,-2} [m_2, \tilde{\mu}, m_q^{i1}] - \\
& \frac{5}{9} g_1 m_3 g_3^2 a_u^{i1i2} \text{LF}_{2,1,1,0} [m_3, m_{\bar{q}}^{i1}, m_{\bar{u}}^{i2}] + \frac{1}{6} g_1 m_3 g_3^2 a_u^{i1i2} \text{LF}_{2,2,1,-1} [m_3, m_q^{i1}, m_{\bar{u}}^{i2}] + \\
& \frac{5}{9} g_1 m_3 g_3^2 a_u^{i1i2} \text{LF}_{3,1,1,-1} [m_3, m_q^{i1}, m_{\bar{u}}^{i2}] - \frac{1}{6} g_1 m_3 g_3^2 a_u^{i1i2} \text{LF}_{2,2,1,-1} [m_3, m_{\bar{u}}^{i2}, m_q^{i1}] + \\
& \frac{1}{16} g_1 \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{i1r} y_u^{pi2} \text{LF}_{2,2,1,-1} [m_d^r, \tilde{\mu}, m_q^p] - \\
& \frac{1}{16} g_1 \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{i1r} y_u^{pi2} \text{LF}_{2,2,1,-1} [m_q^p, \tilde{\mu}, m_d^r] + \frac{5}{72} g_1^3 y_u^{i1i2} \text{LF}_{2,1,1,-1} [\tilde{\mu}, m_1, m_q^{i1}] - \\
& \frac{1}{18} g_1^3 y_u^{i1i2} \text{LF}_{3,1,1,-2} [\tilde{\mu}, m_1, m_q^{i1}] - \frac{5}{18} g_1^3 y_u^{i1i2} \text{LF}_{2,1,1,-1} [\tilde{\mu}, m_1, m_{\bar{u}}^{i2}] + \\
& \frac{1}{18} g_1^3 y_u^{i1i2} \text{LF}_{3,1,1,-2} [\tilde{\mu}, m_1, m_{\bar{u}}^{i2}] - \frac{5}{8} g_1 g_2^2 y_u^{i1i2} \text{LF}_{2,1,1,-1} [\tilde{\mu}, m_2, m_q^{i1}] + \\
& \frac{1}{2} g_1 g_2^2 y_u^{i1i2} \text{LF}_{3,1,1,-2} [\tilde{\mu}, m_2, m_q^{i1}] + \frac{1}{24} g_1 \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{i1r} y_u^{pi2} \text{LF}_{2,1,1,0} [\tilde{\mu}, m_d^r, m_q^p] + \\
& \left. \frac{5}{24} g_1 \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{i1r} y_u^{pi2} \text{LF}_{3,1,1,-1} [\tilde{\mu}, m_d^r, m_q^p] \right)
\end{aligned}$$